IPSAS School Finance System

INTERNAL TECHNICAL DOCUMENT

Gap Analysis & Implementation Roadmap

A comprehensive review of the current system state against IPSAS compliance requirements, with prioritised fixes and a phased implementation plan.

DATE	SYSTEM	STACK	STATUS
April 2026	IPSAS Finance v2	Next.js 15 · Prisma · PostgreSQL	Active Development

SECTION 1

Executive Summary

This document presents a dual-source gap analysis: a manual review of the attached PDF specification and an independent code inspection of the live repository. The findings are reconciled, deduplicated, and prioritised into a phased implementation plan.

15	5	8	4
FEATURES COMPLETE	PARTIALLY DONE	GAPS IDENTIFIED	CRITICAL / HIGH PRIORITY

Methodology: The PDF specification (13 pages) was compared against a full code inspection of 47 API routes, 6 service files, the Prisma schema (38 models), and all dashboard pages. Each claim in the PDF was verified against actual running code, not assumptions.

Key Findings at a Glance

Domain	PDF Assessment	Code Reality	Status
Financial Statements (FP, FPerf, CF)	Partially reliable; needs validation	Logic is sound; relies on StatementLine seed completeness	Conditional
Trial Balance	Present	Missing POSTED-only filter — includes draft vouchers	Bug
General Ledger	Needs consolidated view	Per-account only; no cross-account consolidated view	Partial
Budget vs Actuals	Missing	Implemented & fixed (debitLc/glHeader)	Done
Departmental Expenditure	Missing	Implemented & fixed	Done
AR / AP Ageing	Present	Currency filter excludes cross-currency invoices in totals	Partial
Stock Take	Missing UX; DB tables absent	Schema, API & full UX rebuilt this session	Done
Inventory Variance Report	Required	No dedicated page or API exists	Missing
Stock Requisition → Issue → Movement	Workflow present	Issue action does not create InventoryMovement record	Bug

Domain	PDF Assessment	Code Reality	Status
Voucher Reject Route	Missing	Added this session	Done
Exceptions & Audit Report	Required	Full ExceptionReportService with 7 sections	Done
Multi-Currency Integrity	Needs discipline	debitLc/creditLc vs debitFc/creditFc correctly handled in reports	Partial
Demo Data Realism	Needs enrichment	Minimal seed — flat charts, limited transactions	Thin

SECTION 2

Confirmed Working Features

Features below were verified against source code. Each is production-ready or requires only minor data-seeding to activate.

Financial Reporting

Feature	Key File(s)	Notes	
Financial Position (Balance Sheet)	`report.service.ts:518`	StatementLine tree + account hierarchy + currency switching	✓
Financial Performance (Income Statement)	`report.service.ts:522`	Revenue/Expense summary, surplus calculation, margin %	✓
Cash Flow Statement	`report.service.ts:526`	Operating/Investing/Financing via CF-100/200/300 codes	✓
Trial Balance	`report.service.ts:8`	groupBy accountId; debitLc/creditLc — *missing POSTED filter*	△
General Ledger (per account)	`report.service.ts:72`	Paginated, opening balance, running balance, charts, Lc/Fc	✓
Budget vs Actuals	`api/reports/budget-vs-actuals`	Fixed: glHeader relation + debitLc/creditLc; chart by account type	✓
Departmental Expenditure	`api/reports/departmental-expenditure`	Cost centre grouping; fund dimension; top-8 chart	✓
AR Ageing	`report.service.ts:530`	Buckets: Current / 31–60 / 61–90 / 91–120 / 121+ days	✓
AP Ageing	`report.service.ts:548`	Same bucket logic for supplier bills	✓
Exceptions & Audit Report	`exception-report.service.ts`	7 sections: backdated, reopened, missing attachments, journals, overrides, reversals	✓
Audit Log	`report.service.ts:299`	Full filter by user, entity, action, date range	✓

Accounting Workflows

Feature	Key File(s)	Notes	
Manual Journals (GL Vouchers)	`api/vouchers`	Create, submit, approve, post, reject lifecycle complete	✓
AR Invoice → Post → GL	`api/ar/invoices`	Voucher created on post; debitFc/creditFc written for USD	✓
AR Receipt / Payment	`api/ar/receipts`	Balance deducted from invoice on allocation	✓
AP Bill → Post → GL	`api/ap/bills`	Supplier payable posted to GL on approve/post	✓
AP Payment	`api/ap/payments`	Clears supplier bill balance	✓
Bulk Invoice Generation	`api/ar/invoices/bulk-generate`	Fee templates → batch invoice for all students	✓
Procurement (PO → GRN → AP Bill)	`api/procurement`	3-way matching implemented	✓
Fixed Asset Register	`api/assets`	Categories, depreciation run, pending from GRN	✓

Inventory & Stock

Feature	Key File(s)	Notes	
Inventory Items & Categories	`api/inventory/items`	Full CRUD; linked to GL account	✓
Inventory Receipts	`dashboard/inventory/receipts`	GRN-linked receipts increase stock balance	✓
Stock Requisitions	`api/inventory/requisitions`	Draft → Submit → Approve flow; issue step present	△
Stock Take (Session + Count)	`api/inventory/stock-take`	Rebuilt this session — DRAFT→COUNTED→APPROVED→POSTED	✓

SECTION 3

Detailed Gap Analysis

Each gap below is documented with its root cause, affected files, risk level, and recommended fix. Items are ordered by priority.

1 Trial Balance Includes Unposted Vouchers

CRITICAL | Data Integrity | 5-Minute Fix

The `getTrialBalance()` method queries **all GL entries** without filtering for `voucher.status = "POSTED"`. This means draft, pending, and rejected vouchers inflate the Trial Balance figures, leading to financial misstatement.

Affected File `src/lib/services/report.service.ts` — lines 14–31

Root Cause The `glHeader` where clause only filters by `organisationId` and `entryDate`. No `voucher: { status: "POSTED" }` condition present.

Risk Trial Balance will not match Financial Statements (which use the same GLEntry table correctly in getFinancialStatement). Auditors will immediately notice the discrepancy.

Fix Add `voucher: { status: "POSTED" }` to the `glHeader` filter in `getTrialBalance()`. One line change.

> **Note:** This is also missing from `getTrialBalance()` but correctly present in `getFinancialStatement()` — confirming this is an oversight, not an architectural decision.

2 Financial Statements Silently Return Zero if Statement Lines Not Seeded

HIGH | Reliability | Setup Dependency

`getFinancialStatement()` relies entirely on the `StatementLine` and `AccountStatementMap` tables being correctly seeded. If a new organisation is created without running the COA seed, all three financial statements (Financial Position, Financial Performance, Cash Flow) return empty rows with zero totals — no error, no warning.

Affected Files `report.service.ts:326–515`, all three financial statement pages

Root Cause Architecture is correct — tree-based statement lines are powerful. But there is no validation or warning when the seed is absent.

Risk A user in a demo or new environment sees all-zero reports and assumes the system is broken or that no transactions exist.

Fix Add a `_meta.mappedAccountCount` field to the response. If 0, the API returns HTTP 200 with a `warning: "No statement lines configured"` message. The UI surfaces this with a setup prompt.

Stock Requisition Issue Does Not Create Inventory Movement

HIGH | Data Integrity | Inventory

When a stock requisition is fulfilled (status → FULLY_ISSUED), the system updates `qtyIssued` on the requisition lines but **does not** call `InventoryMovementService` to decrement the item's on-hand quantity. The stock balance never decreases, making the inventory count permanently incorrect.

Affected File `src/app/api/inventory/requisitions/[id]/route.ts` — issue action handler

Root Cause The issue workflow was implemented at the requisition level without wiring up the downstream inventory ledger write. Two separate concerns were not connected.

Risk Inventory reports, stock take variances, and reorder calculations are all based on stale quantity data. The stock take will always show large positive variances because the system quantity is never decremented.

Fix After updating `qtyIssued`, call `prisma.inventoryMovement.create({ type: "ISSUE", quantity: -qty, ... })` for each requisition line. Update the item's `quantity` field accordingly.

4 Inventory Variance Report — Page and API Missing

MEDIUM | Feature Gap | 2–3 Hours

Stock takes record variances (physical qty vs system qty) but there is no dedicated report to view these variances across sessions, items, or time periods. The variance data sits in `stock_take_lines` but is inaccessible from the UI without opening each session individually.

Missing Files `src/app/dashboard/inventory/variance-report/page.tsx`, `src/app/api/inventory/variance-report/route.ts`

Required Data Join `StockTakeLine → InventoryItem → StockTakeSession`; filter where `variance ≠ 0`; calculate cost impact using item's average cost.

PDF Requirement The attached spec explicitly lists "Variance Report" and "Value of Variance" as required outputs from the stock governance module.

5 AR/AP Ageing: Multi-Currency FX Not Converted

MEDIUM | Multi-Currency | 3–4 Hours

`getARAgeing()` and `getAPAgeing()` pass `currencyCode: options.reportingCurrency` directly into the Prisma `where` clause. This excludes invoices in other currencies from the totals entirely. A request for ZWG totals will silently omit all USD invoices.

Affected File `report.service.ts:530–564`

Correct Behaviour All invoices should appear regardless of currency. If the reporting currency differs from the invoice currency, the balance should be converted using the voucher's `fxRate`.

Fix Remove the `currencyCode` filter from the `where` clause. Include the invoice's `currencyCode` and `voucher.fxRate` in the query. In `calculateAgeing()`, multiply `balance` by `fxRate` when currencies differ.

6 Consolidated General Ledger (All Accounts) Missing

MEDIUM | Feature Gap | 2–3 Hours

The current General Ledger report is account-specific: users must select an account before viewing entries. The PDF specification requires a consolidated view showing all accounts' movements in a date range, filterable by source module, account type, fund, and cost centre — similar to a full journal listing.

Current Behaviour	`GET /api/reports/general-ledger?accountId=xxx` — requires a specific account ID
Required Behaviour	`GET /api/reports/general-ledger` (no accountId) — returns all entries across all accounts, paginated, with filter params
Fix	Make `accountId` optional in `getGeneralLedger()`. When absent, omit the `accountId` filter and group/sort by account + date. Add `sourceModule` and `accountType` filter params.

7 Stock Take Not Linked in Sidebar Navigation

LOW | Navigation | 5-Minute Fix

The Stock Take page exists at `/dashboard/inventory/stock-take` but there is no sidebar link pointing to it. Users can only reach it by typing the URL directly or navigating from the Inventory overview page — if that link exists.

Affected File	`src/components/layout/app-sidebar.tsx` — Inventory sub-section
Fix	Add `{ title: "Stock Take", href: "/dashboard/inventory/stock-take", icon: ClipboardCheck }` to the Inventory links array.

8 Demo Data Too Thin for Meaningful Report Previews

LOW | Demo Quality | 3–4 Hours

The current seed file produces minimal data: a few invoices, one or two suppliers, and no historical GL entries. All charts appear flat or empty, which weakens the demo story. The PDF specification notes the need for realistic school-context data.

Affected File	`prisma/seed.ts`
What's Needed	24 months of monthly journal entries (salary, utilities, school fees), 20+ inventory movements, 5+ suppliers with bill/payment history, 50+ student fee invoices across 3 terms

SECTION 4

Implementation Roadmap

Gaps are grouped into four phases based on urgency, data integrity risk, and interdependencies. Each phase is independently deployable.

PHASE 1 — IMMEDIATE

Data Integrity Fixes

- → Fix Trial Balance POSTED filter
- → Add Stock Take sidebar link
- → Add balance sheet equation check

PHASE 2 — THIS WEEK

Stock Integrity

- → Requisition issue → InventoryMovement
- → Inventory Variance Report page + API
- → Statement line setup warning in UI

PHASE 3 — NEXT WEEK

Reporting Completeness

- → Consolidated GL (all accounts view)
- → AR/AP Ageing FX conversion
- → Supplier & Student Statement reports

PHASE 4 — POLISH

Demo Readiness

- → Rich seed data (24-month history)
- → Multi-currency demo transactions
- → Onboarding wizard for new orgs

Phase 1 — Detailed Tasks (Immediate)

#	Task	File	Change	Effort
1.1	Trial Balance POSTED filter	`report.service.ts:14`	Add `voucher: { status: "POSTED" }` to glHeader where	5 min
1.2	Stock Take sidebar link	`app-sidebar.tsx`	Add ClipboardCheck link to Inventory section	5 min
1.3	Balance sheet equation check	`report.service.ts:429–438`	Compute `isBalanced = |assets - liab - netAssets| < 0.01`	20 min

Phase 2 — Detailed Tasks (This Week)

#	Task	File	Change	Effort
2.1	Requisition issue → movement	`api/inventory/requisitions/[id]/route.ts`	After qtyIssued update, create InventoryMovement ISSUE record; decrement item.quantity	2 hrs
2.2	Variance Report API	`api/inventory/variance-report/route.ts (new)`	Query stock_take_lines where variance≠0; join item+session; calculate cost impact	1 hr
2.3	Variance Report UI	`dashboard/inventory/variance-report/page.tsx (new)`	Table: Item / Session / Date / System Qty /	1.5 hrs

#	Task	File	Change	Effort
			Physical Qty / Variance / Value Impact	
2.4	Statement line warning	`report.service.ts:326 + financial-position page`	If mappedAccountCount=0, add warning field to response; show setup banner in UI	1 hr

Phase 3 — Detailed Tasks (Next Week)

#	Task	File	Change	Effort
3.1	Consolidated GL (all accounts)	`report.service.ts:72 + general-ledger route`	Make accountId optional; when absent, fetch all entries with account grouping	2 hrs
3.2	AR/AP Ageing FX conversion	`report.service.ts:530–564`	Remove currencyCode filter; include fxRate; convert balance in calculateAgeing()	3 hrs
3.3	Supplier Statement report	`api/reports/supplier-statement/route.ts (new)`	Per-supplier: bills, payments, running balance, date range filter	2 hrs
3.4	Student Statement report	`api/reports/student-statement/route.ts (new)`	Per-student: invoices, receipts, running balance, term filter	2 hrs

Phase 4 — Demo Data Enrichment

#	Task	Detail	Effort
4.1	Historical GL journals (24 months)	Monthly: salaries, utilities, school fees, maintenance — across all account types	2 hrs
4.2	Inventory movement history	20+ receipts and issues across 10 items over 12 months	1 hr
4.3	Multi-currency transactions	5 USD invoices + 5 USD bills with FX rates to demonstrate currency switching	1 hr
4.4	Richer supplier/student data	8 suppliers (cleaning, catering, IT, maintenance), 50 students across 3 classes	1 hr

SECTION 5

Technical Architecture Notes

Key architectural patterns and invariants that must be respected when implementing the fixes above.

GLEntry Field Naming Convention

This is the most common source of bugs in the codebase. All GL entries use a four-field currency model:

Field	Meaning	When to Use
`debitLc` / `creditLc`	Local (base) currency amount — e.g. ZWG	Default for all domestic reports. Use for Trial Balance, Budget vs Actuals, Departmental Expenditure
`debitFc` / `creditFc`	Foreign currency amount — e.g. USD	Use when `reportingCurrency === "USD"` or for USD bank account GL reports
`fxRate`	Exchange rate at posting time (LC per 1 FC)	Use when converting between currencies in ageing reports

Never use `debit` or `credit` — these fields do not exist on the `GLEntry` model. Using them will cause a TypeScript error or silent `undefined` values in JavaScript contexts.

GLEntry Filtering Pattern

GLEntry does not have a direct `organisationId` field. All organisation scoping must go through the `glHeader` relation:

```
// ✓ Correct pattern const entries = await prisma.gLEntry.findMany({ where: { glHeader: {
organisationId, entryDate: { gte: startDate, lte: endDate }, voucher: { status: "POSTED" }
// Always include this! } } }); // ✗ Wrong — organisationId does not exist directly on
GLEntry const entries = await prisma.gLEntry.findMany({ where: { organisationId } //
TypeScript error or runtime failure });
```

Auth Patterns

Pattern	Usage	Returns
`withAuth(req, handler, roles[])`	Older routes; role-gated middleware	Injects `authReq.user.{ userId, organisationId, role }`
`verifyAuth(req)`	Newer report routes	Returns `{ userId, organisationId, role }` or null

StockTakeStatus Lifecycle

DRAFT → COUNTED → APPROVED → POSTED

- **DRAFT:** Session created, lines added, physical counts entered. Editable.
- **COUNTED:** Counts saved, variances calculated. Locked from editing.
- **APPROVED:** Approved by ADMIN/BURSAR/HEADMASTER. Ready to post.
- **POSTED:** Variance adjustments written to InventoryMovement. Immutable.

Multi-Currency Rule Summary

✓ GL reports use `useFc = reportingCurrency === "USD"` flag to pick Lc vs Fc fields

✓ Financial statements correctly pass `reportingCurrency` through the service chain

✓ Budget vs Actuals always uses `debitLc` (local currency base)

△ **AR/AP Ageing excludes foreign-currency invoices from totals when currency filter is applied**

✗ Consolidated GL does not exist yet — multi-currency support to be added alongside it

✗ No FX rate conversion utility function exists — each service handles it ad-hoc

Prisma Migration Notes

> This project uses **`prisma db push`** (schema push) rather than migrations. This means there is no migration history file. For destructive operations (dropping columns, renaming), `prisma db push --force-reset` is required with the `PRISMA_USER_CONSENT_FOR_DANGEROUS_AI_ACTION=true` environment variable set. Always re-seed after a force reset.

Seed Credentials (Demo Environment)

Role	Email	Password
Admin	`admin@school.ac.zw`	`Admin@123`
Bursar	`bursar@school.ac.zw`	`Bursar@123`
Teacher	`teacher@school.ac.zw`	`Teacher@123`